Experiment 03 – Lab 6

Document the Prompt Design and Output Evaluation

Aim

To design clear and specific prompts for ChatGPT, record the outputs, and evaluate them for relevance, accuracy, completeness, and format.

Theory

A prompt is the instruction or question given to an AI system. A well-designed prompt states the task clearly, provides enough context, identifies the required format, and includes any important constraints. Output evaluation checks whether the response follows the prompt and meets the intended requirements.

Procedure

1. Select a simple task.
2. Write a basic prompt and note the output.
3. Improve the prompt by adding context, constraints, and an output format.
4. Compare the responses using an evaluation checklist.
5. Record the findings and conclusion.

Task 1: Basic Prompt

Prompt: Explain photosynthesis.

Sample output: Photosynthesis is the process by which green plants use sunlight, water, and carbon dioxide to produce glucose and release oxygen.

Evaluation: The answer is relevant and broadly accurate, but a beginner may benefit from a simpler explanation and a step-by-step format.

Task 2: Improved Prompt

Prompt: Explain photosynthesis to a beginner in simple English. Use exactly three numbered points and include the inputs and outputs of the process.

Sample output:
1. Plants absorb sunlight using chlorophyll in their leaves.
2. They take in water from the soil and carbon dioxide from the air.
3. Using light energy, they produce glucose (food) and release oxygen.

Evaluation: The improved answer is easier to follow, has the requested three-point format, and includes the main inputs and outputs. It is more aligned with the prompt.

Output Evaluation Table

Criterion	Basic Prompt Output	Improved Prompt Output
Relevance	Relevant to the topic	Relevant and tailored to a beginner
Accuracy	Broadly correct for a short definition	Broadly correct; includes key inputs and outputs
Completeness	Short overview only	Covers sunlight, water, carbon dioxide, glucose, and oxygen
Format	No format specified	Exactly three numbered points
Clarity	Understandable but brief	Simple language and organized steps

Findings

Adding the audience, required details, and output structure improved the usefulness of the response. The improved prompt made the expected level of detail and format explicit, making the answer easier to assess.

Best Practices for Prompt Design

- State one clear task.
- Specify the audience and context.
- Mention important details or constraints.
- Request a suitable format, such as bullets, a table, or numbered steps.
- Check factual claims and verify that all requirements are met.

Conclusion

Prompt design has a direct effect on the relevance, clarity, completeness, and format of ChatGPT's responses. Evaluating outputs against explicit criteria helps determine whether the response satisfies the task.

Result

Successfully documented two prompt designs and evaluated their outputs. The improved prompt produced a more structured response that better met the stated requirements.

Note: The responses are illustrative sample outputs. ChatGPT may phrase answers differently across runs.